Assignment 3

Introduction

This paper is divided into two parts. Part 1 aims to answer the first research question:

> *RQ1: Are there any significant associations between the personality of students preferred animal?*

For this research question a group of students were classified in terms of two personality types: Introvert and Extrovert. The students were also asked which animal they preferred among four options: tiger, bat, rabbit, or whale. The data was then put into a txt file that I was provided by the course instructor with the intention to perform a Chi-Square test with RStudio to answer the research question posed.

I am very interested in videogames, and I spend my free time watching e-sport matches, especially *Valorant* and *Counter Strike*. One fact that has always impressed me is the prize earnings that teams can earn if they win a tournament. This made me wonder if prize earnings depend on the game genre. Thus, Part 2 of this paper aims to answer the following research question:

> *RQ2: Are there any significant associations between e-sports prize earnings and a team's game genre?*

To answer this question, I used a dataset containing information on 2024 prize earnings by teams amongst 5 different game genres. The dataset was cleaned, and ultimately imported into RStudio to conduct a Chi-Square test.

This paper was written in Rmarkdown, knitted, and then slightly formatted as a Word Document.

Methods

a. Part 1

The dataset used consists of data provided by the *Statistical analyses and visualization in R:I* course instructors as a txt file. The dataset consists of students, their personality type (introvert or

extrovert), and their preferred animal (tiger, bat, rabbit, or whale). The data was analyzed with Rstudio, and online sources were used as support.

To answer the research question *Are there significant associations between the personality of students and preferred animal?* I used Rstudio to perform a Chi-Square test to test if there are any relationships between the students' personality and preferred animal. A Chi-Square is fitting given that I am exploring the association between two categorical variables. The null hypothesis H0: There is no relationship between a student's personality and their preferred animal.

The first step was to make sure to remove any missing values with the code below. The cleaned data has the same number of observations, and variables, which tells me that there were no missing values to remove.

```
clean_df <- na.omit(df)
```

To be able to run the Chi-Square test, I had to create a contingency table (or a matrix) using the code below.

```
Contingency.Table <- table("rows" = df$Personality, "columns" = df$Animal)

Contingency.Table
```

Once the contingency table was ready, I performed the Chi-Square test in which the null hypothesis (H0) states that there is no association between personality and preferred animal,. On the other hand, the alternative hypothesis (H1) states that there is a significant association between personality and preferred animal. I also indicated the critical value according to 3 degrees of freedom and 0.05 significance level using an online source.

```
  Chi.Test <- chisq.test(Contingency.Table)
  Chi.Test
```

To get a more complete analysis, I also found the expected frequencies, which are those that we would expect under the assumption that there is no association between the two variables. I also looked into the residuals to identify further patterns. Then, the observed frequencies, expected frequencies, and the residuals from the Chi-Square test were put into Table 3.

```
    # Expected frequencies

expected.frequencies <- Chi.Test$expected
print(round(expected.frequencies, 2))

    # Get raw residuals

raw_residuals <- Chi.Test$observed - Chi.Test$expected

  #Putting observed, expected, and residuals in one table

merged_tables <- cbind(round(raw_residuals), Contingency.Table,
expected.frequencies)

kable(merged_tables, caption = "Table 3. Residuals, Observed Frequencies,
Expected Frequencies") %>%
  kable_classic(full_width = FALSE) %>%
 add_header_above(c(" " = 1, "Residuals" = 4, "Observed" = 4, "Expected" = 4))
%>%
  row_spec(0, italic = TRUE) %>%
  column_spec(2:5, background = "mistyrose") %>%
  column_spec(6:9, background = "ivory") %>%
  column_spec(10:13, background = "azure") %>%
  column_spec(1, border_right = TRUE) %>%
  column_spec(5, border_right = TRUE) %>%
  column_spec(9, border_right = TRUE)
```

To get a better visualization of the data, I created a barplot showing the same data as the contingency table - showing frequencies. I also created a heatmap of the squared standardized residuals to see where the Chi-Square statistics are being driven.

```
 # Barplot

barplot(Contingency.Table,
        beside=T,
        col=c("indianred3", "steelblue4"),
        xlab="Preferred Animal",
```

```
        ylab="Frequency",
        main="Figure 1. Personality and Preferred Animal",
        legend.text=c("Extrovert", "Introvert"),
        args.legend = list(x="topright", bty="n"),
        cex.axis = 0.75,
        cex.names = 0.7)

  # Creating a heatmap of squared standardized residuals

pheatmap(round(Chi.Test$residuals^2, 2),
         display_numbers = TRUE,
         cluster_rows = FALSE,
         cluster_cols = FALSE,
         main = "Figure 3. Chi-Square Residuals")
```

b. Part 2

The dataset used for this part was downloaded from Kaggle as a zip file (link to data provided in the references). The zip-file contained 4 spreadsheets which were extracted and manually browsed to select the two sheets that contained the data that would be relevant to answer the research question.

I also used Excel to correct the county's abreviation into the actual country name. I ended with one Excel document with one sheet listing the *Player's ID*, *First* and *Last Name*, the players *in-game name*, *Country*, *Continent*, *Total USD Prize*, *Game*, and *Genre*. The variables I used were *Total USD Prize* measured in dollars, and *Genre*, categorizing a specific game as one of the following 5: *Battle Royale*, *Collectible Card Game*, *First-Person Shooter*, *Multiplayer Online Battle Arena* and *Strategy*.

The Excel sheet was imported into Rstudio to be analyzed.

```
library(readxl)

esports_earnings <- read_excel("C:/Users/angel/OneDrive/Desktop/School/R
Course/Assignment 3/esports_earnings.xlsx")

View(esports_earnings)
```

```
table(esports_earnings$Genre)
```

I first wanted to see the distribution of the team earnings to get an idea of the data. For that I created a histogram:

```
hist(esports_earnings$TotalUSDPrize / 1e6,
     col = "orchid",
     border = "black",
     main = "Figure 3. e-sport Teams and Total USD Prize Money",
     xlab = "Prize Earnings in Millions (USD)",
     ylab= "Frequency"
     )
```

The first step to be able to answer the research question was to turn the variable *TotalUSDPrize* into a categorical variable with different groups. To do so I used the following code:

```
   # Finding how I should split the categories by looking at measures of
central  tendency

min(esports_earnings$TotalUSDPrize)
max(esports_earnings$TotalUSDPrize)
mean(esports_earnings$TotalUSDPrize)
median(esports_earnings$TotalUSDPrize)

breaks <- c(0, 50000, 150000, 400000, 1000000, 7000000)
labels <- c("Very Low", "Low", "Middle", "High", "Very High")

esports_earnings$EarningsCategory <- cut(esports_earnings$TotalUSDPrize,
                                  breaks = breaks,
                                  labels = labels,
                                  right = TRUE,
                                  include.lowest = TRUE
                                  )

table(esports_earnings$Genre)
```

Once I had my categorical variable *EarningCategory*, I created a contingency table for the *Team Earnings Category* and *Genre*.

```
Contingency.Table2 <- table( "rows" = esports_earnings$Genre, "columns" =
esports_earnings$EarningsCategory)
```

Once the contingency table was ready, similar to Part 1, I performed the Chi-Square test in which the null hypothesis (H0) states that there is no association between price earnings and e-sport genre. On the other hand, the alternative hypothesis (H1) states that there is a significant association between e-sport price earnings and game genre.

```
Chi.Test2 <- chisq.test(Contingency.Table2)

Chi.Test2
```

Similar to Part 1, I found the expected frequencies and residuals to get a deeper analysis. The observed frequencies, expected frequencies, and the residuals from the Chi-Square test have been put into Table 5.

```
    # Expected frequencies

expected.frequencies2 <- Chi.Test2$expected

print(round(expected.frequencies2, 2))

    # Get raw residuals

raw_residuals2 <- Chi.Test2$observed - Chi.Test2$expected

    #Putting observed, expected, and residuals in one table

merged_tables<-cbind(round(raw_residuals2), Contingency.Table2,
round(expected.frequencies2))

kable(merged_tables, caption = "Table 3. Residuals, Observed Frequencies,
Expected Frequencies") %>%
  kable_classic(full_width = FALSE) %>%
  add_header_above(c(" " = 1, "Residuals" = 5, "Observed" = 5, "Expected" = 5))
%>%
  row_spec(0, italic = TRUE) %>%
  column_spec(2:6, background = "cornsilk") %>%
```

```
  column_spec(7:11, background = "thistle1") %>%
  column_spec(12:16, background = "lightcyan") %>%
  column_spec(1, border_right = TRUE) %>%
  column_spec(6, border_right = TRUE) %>%
  column_spec(11, border_right = TRUE)
```

To get a good visualization of the data, I created a bar-plot showing the frequency of different teams within a genre by their earnings category.

```
barplot(Contingency.Table2,
        beside=T,
        col=c("firebrick", "orange", "seagreen", "royalblue3", "hotpink3"),
        xlab="Prize Earnings Category",
        ylab="Frequency",
        main="Figure 4. Game Genre and Teams Prize Earnings",
        legend.text=c("Battle Royale", "Collectible Card Game", "First-Person
Shooter","Multiplayer Online Battle Arena", "Strategy"),
        args.legend = list(x="topright", bty="n"),
        cex.axis=0.75,
        cex.names=0.7
        )
```

The residuals were also plotted in a heat map to show where the Chi-Square statistic is being driven.

```
pheatmap(round(Chi.Test2$residuals^2, 2),
         display_numbers = TRUE,
         cluster_rows = FALSE,
         cluster_cols = FALSE,
         main = "Figure 5. e-sports Squared standardized Residuals",
         color = colorRampPalette(c("white", "lightblue", "royalblue"))(100),
         number_color = "black")
```

Results & Analysis

a. Part 1

Table 1. Contingency Table Personality and Preferred Animal

	Tiger	Bat	Rabbit	Whale
Extrovert	20	16	13	10
Introvert	2	11	18	10

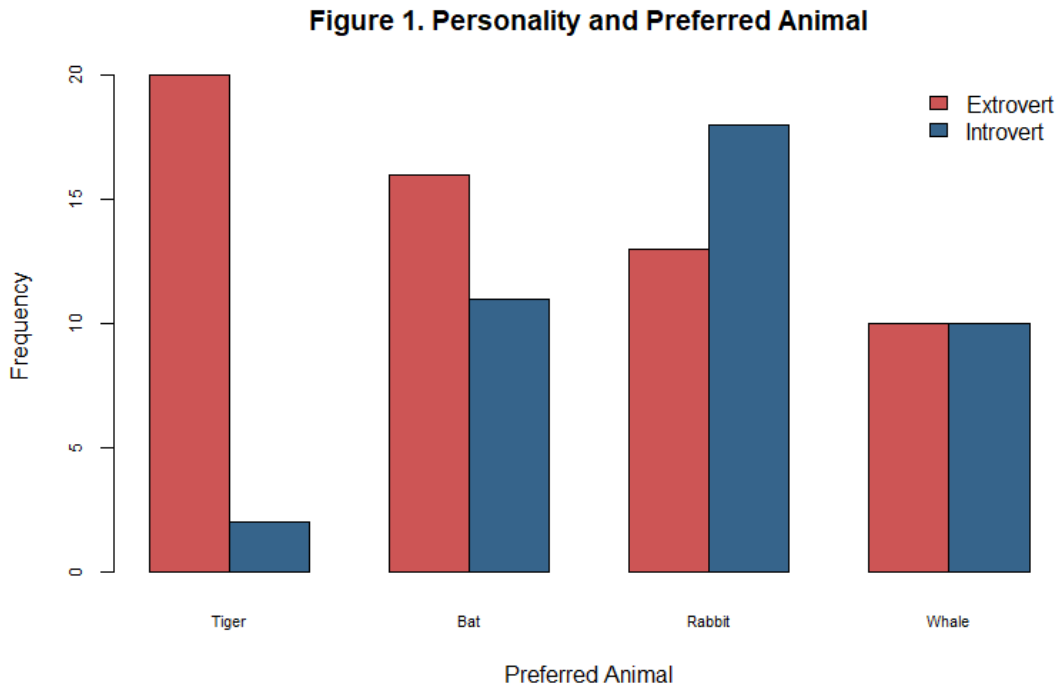


Figure 1. Frequency of Personality and Preferred Animal

Table 2. Chi-Square Results Personality & Animal

H0: There is no association between the two variables.

H1: There is a significant association between the two variables.

Critical Value: 7.815

X-Squared	**df**	**P-value**

13.662	3	0.003403

The Chi-Square value of 13.662 is greater than the critical value (7.815), suggesting a statistically significant difference between the observed and expected values. The P-value of the Chi-Square tests is smaller than the 0.05 significance level, therefore, we reject the null hypothesis. Further, there's a 0.34% chance the difference between the observed and the expected values happened due to randomness.

Table 3. Residuals, Observed Frequencies, Expected Frequencies

	Residuals				Observed				Expected			
	Tiger	*Bat*	*Rabbit*	*Whale*	*Tiger*	*Bat*	*Rabbit*	*Whale*	*Tiger*	*Bat*	*Rabbit*	*Whale*
Extrovert	7	0	-5	-2	20	16	13	10	12.98	15.93	18.29	11.8
Introvert	-7	0	5	2	2	11	18	10	9.02	11.07	12.71	8.2

Although the Chi-Square test indicates significant association ($P<0.05$), the residuals give further insight on the data. Positive residuals indicate that the observed value is higher than expected, whereas small residuals (close to 0) suggest that the observed value is close to the expected ones. Thus, from Table 3 we can see that Tiger has both high and low residuals, a positive value for extroverts – suggesting a higher number of extroverts found themselves preferring tigers, whereas introverts had -7, suggesting that introverts chose tiger less than the expected number. A similar situation happened with Rabbits but inversely, so extroverts picked Rabbit less than expected, whereas introverts picked Rabbit more than expected.

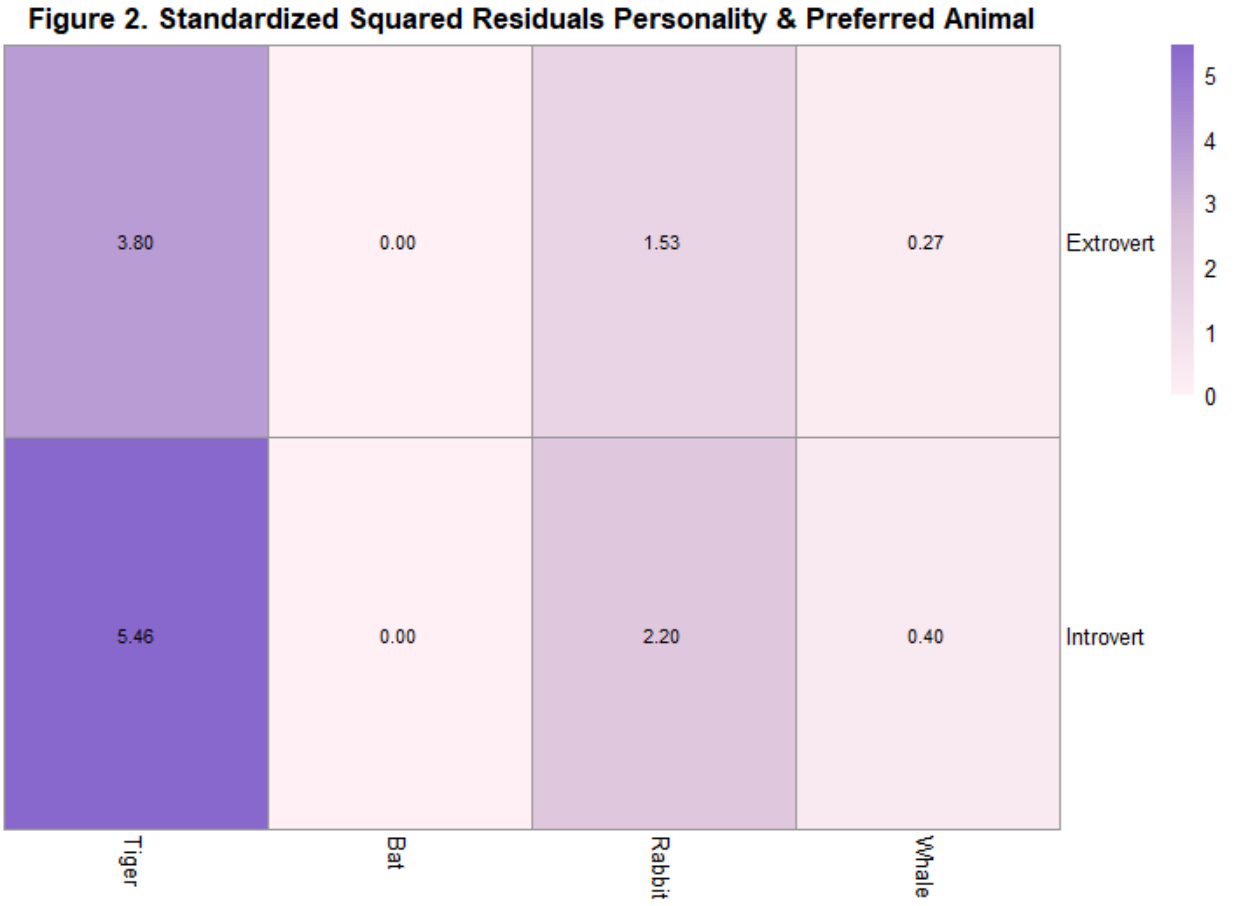


Figure 2. Heatmap of Chi-Square residuals

Figure 2 shows the squared standardized residuals of the contingency table comparing *Personality* and *Preferred Animal*. This heat map allows us to understand which deviate most from their expected counts, meaning which cells impact the Chi-Square statistic the most. For example, our Chi-Square value was 13.662, this means that the Tiger-Introvert cell contributed around 40% to that total, making a big impact on the overall result. Thus, we can say that introverts chose tigers less than expected, strongly influencing the observed association between personality and preferred animal.

b. Part 2

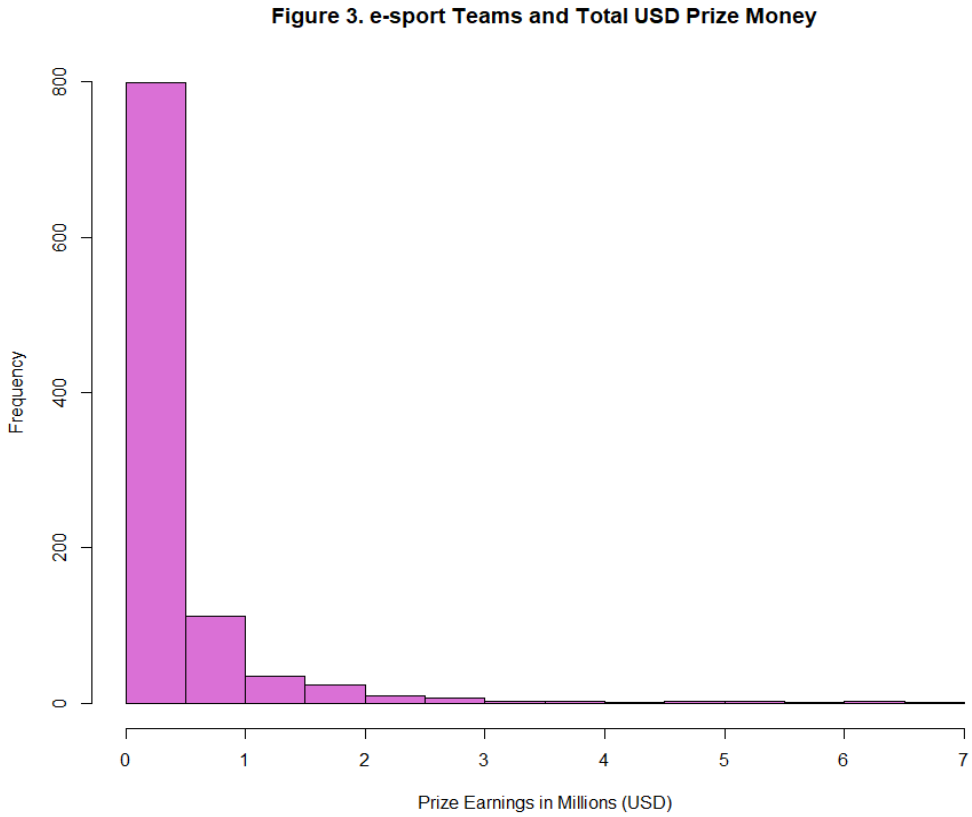


Figure 3. Histogram of e-sport Teams and Prize Earnings

From Figure 3 we can see that the majority of e-sport teams, regardless of game genre, find themselves earning less than 1 million USD in prize money. There are very few teams that have earned more than 3 million. This is significant to the research question, since we want to know if the few teams that earn higher quantities belong to the same game genre.

Table 4. Contingency Table Game Genre and Team Prize Earnings

	Very Low	**Low**	**Middle**	**High**	**Very High**
Battle Royale	8	84	73	22	12
Collectible Card Game	3	66	29	2	0

First-Person Shooter	0	69	75	46	10
Multiplayer Online Battle Arena	41	134	89	70	66
Strategy	0	46	35	19	0

Table 5. Chi-Square Results Prize Earnings & Game Genre

H0: There is no association between the two variables.

H1: There is a significant association between the two variables.

Critical Value: 26.296

X-Squared	**df**	**P-value**
148.15	16	2.2 e-16

For this test, the Chi-Square obtained is 148.15, which is significantly greater than the critical value of 26.296, suggesting a statistically significant difference between the observed and the expected values. Given that the P-value (2.2 e-16) is smaller than the 0.05 significance level, we reject the null hypothesis, pointing to a significant association between Prize Earnings and Game Genre.

Table 6. Residuals, Observed Frequencies, Expected Frequencies

	Residuals					Observed					Expected				
	Very Low	*Low*	*Middle*	*High*	*Very High*	*Very Low*	*Low*	*Middle*	*High*	*Very High*	*Very Low*	*Low*	*Middle*	*High*	*Very High*
Battle Royale	-2	5	13	-10	-6	8	84	73	22	12	10	79	60	32	18
Collectible Card Game	-2	26	-1	-14	-9	3	66	29	2	0	5	40	30	16	9
First-Person Shooter	-10	-11	15	14	-8	0	69	75	46	10	10	80	60	32	18
Multiplayer Online Battle Arena	20	-26	-32	6	31	41	134	89	70	66	21	160	121	64	35
Strategy	-5	6	5	3	-9	0	46	35	19	0	5	40	30	16	9

From Table 6, we can see that there are some residuals that show a higher difference between the observed and expected values. For example, the Multiplayer Online Battle Arena (MOBA) has -26 as a residual for low prize earnings, suggesting that the observed value had a lower frequency (by 26) than the expected value. Differently from the raw residuals in Part 1, larger residuals may

have a lower impact, given that we have larger values. For example, we have to consider that the -26 residual comes from the difference of the expected frequency for MOBA-Low of 160, and the observed value of 134. In this case, the difference is not massive, given that it does not make a high proportion of the expected value. However, we can still say that there are differences between the observed and the expected values of Game Genres and Prize Earnings.

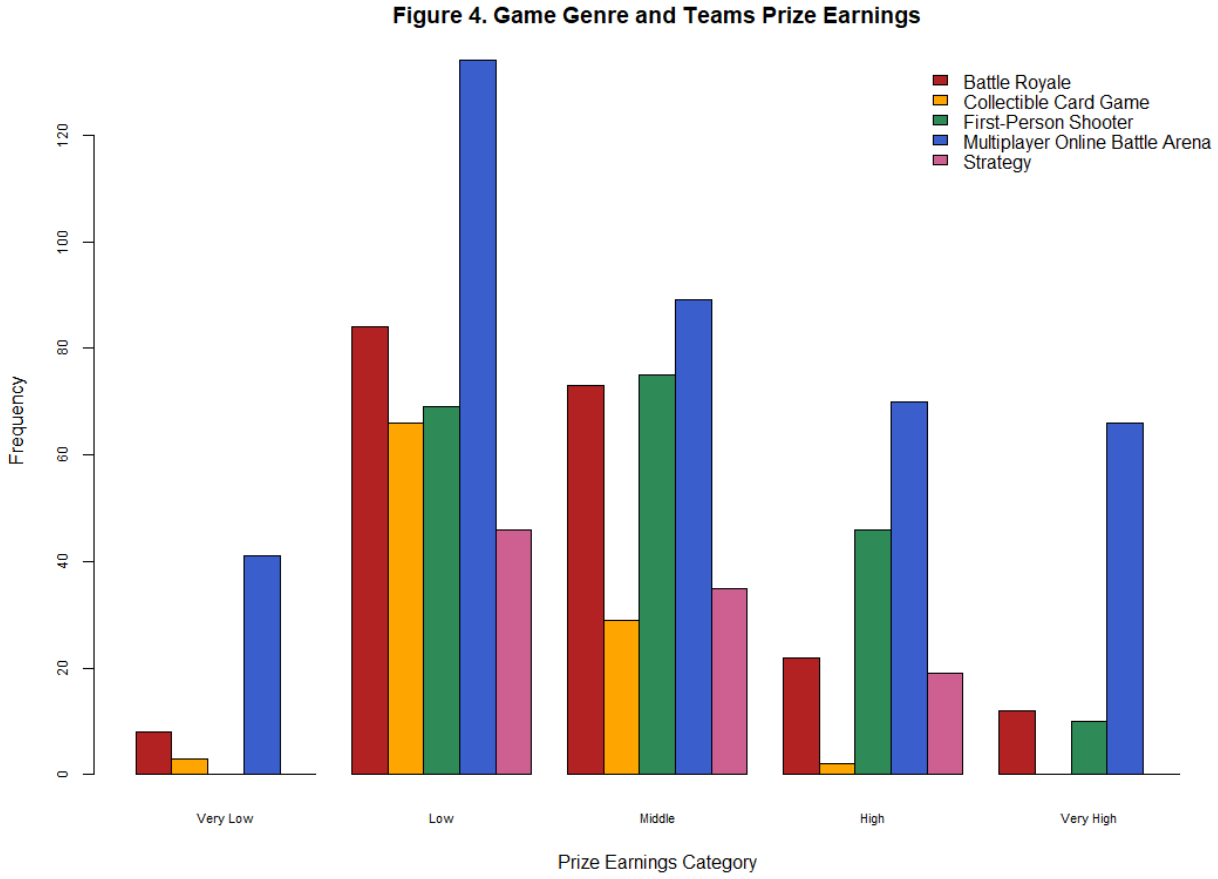


Figure 4. Game Genre and Prize Earnings

We can further see the distribution of teams in the same genre along different prize earnings categories. We can see that MOBA is the most frequent genre amongst all prize earnings, whereas Collectible Card Game are less frequent in high – very high prize earnings.

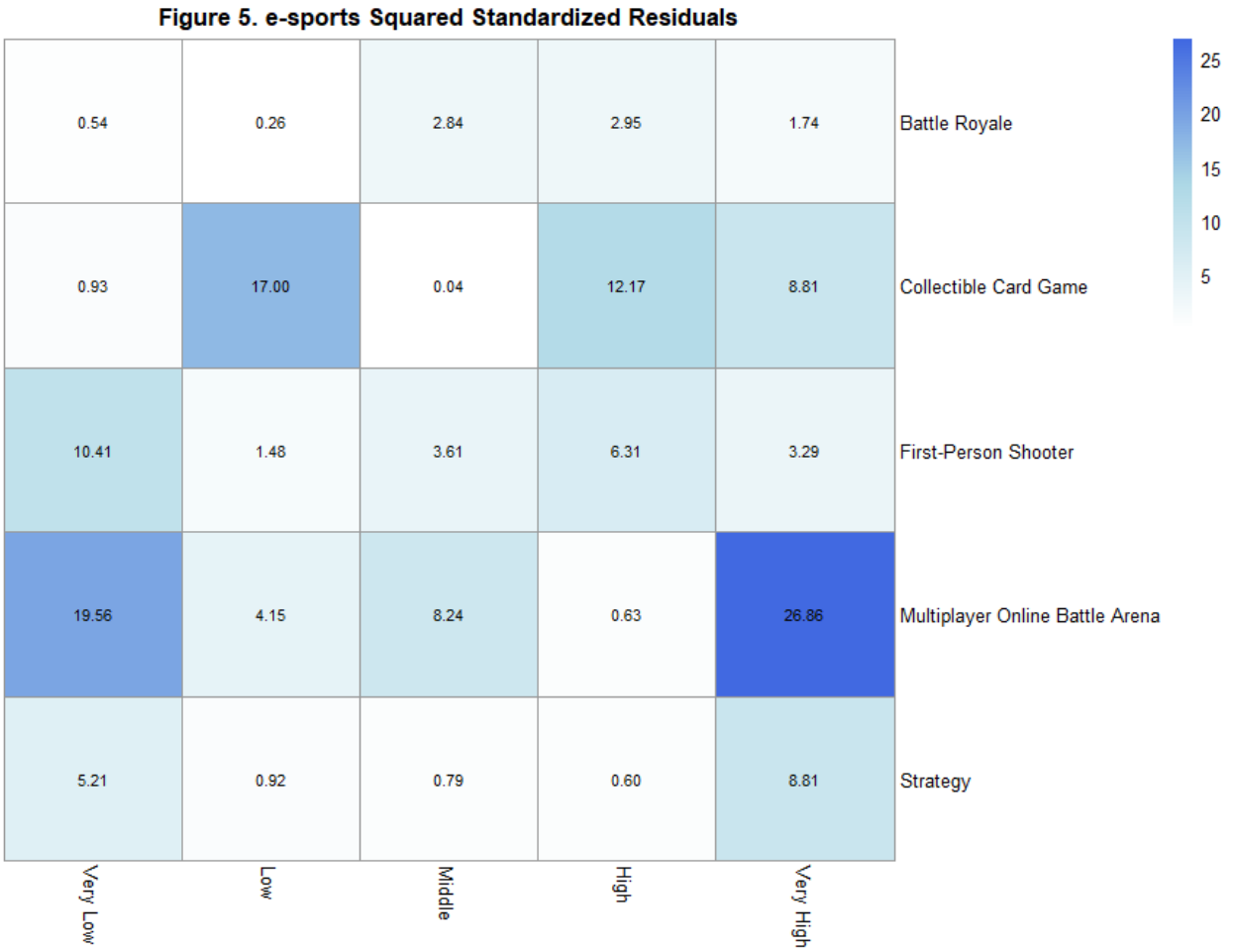


Figure 5. Heatmap of Squared Standardized Residuals - Genre & Team Earnings

Figure 6 shows the squared standardized residuals of the contingency table comparing *Game Genre* and *Prize Earnings*. We can see that most of the Chi-Square value (148.15) is driven by MOBA in both *Very Low* and *Very High* prize earnings (46.42 out of 148.15). However, we can also see that a lot of the other cells still add up to the significance of the Chi-Square value. For example, we can see Collectible Card Game–*Low* adding 17 to the Chi-Square value. In comparison, Battle Royale's have the least impact on the Chi-Square value. Thus, we can say that MOBA teams with very high prize earnings strongly influence the observed associations between game genre and prize earnings.

Discussion

a. Part 1

The results of Part 1 show that there is a significant association between personality type (Extroverted vs Introverted) and the preferred animal answering the research question

> *RQ1: Are there any significant associations between the personality of students preferred animal?*

The Chi-Square value and the P-value from the Chi-Square test of independence support our alternative hypothesis (H1), indicating that there is a significant association between personality and preferred animal and is unlikely to be due to random chance. Further, the residuals revealed notable differences between the observed values and the expected values, suggesting that certain personality types favored/unfavored specific animals more than expected. This association became clearer on Figure 2, where we saw that introvert chose tigers much less than expected – a key driver of the overall association between personality and animal preference. Thus, answering the research question

c. Part 2

Given the results shown in Part 2, we can say that there is a significant association between e-sports team game genres and their prize earnings as per research question

> *RQ2: Are there any significant associations between e-sports prize earnings and a team's game genre?*

The Chi-Square satistics and the P-value from the test of independence support our alternative hypothesis (H1), suggesting that the association is statistically significant and unlikely to be the result of chance. Additionally, the residuals highlighted diferences between the observed and the expected values, indicating that certain game genres tended to earn more or less than expected. This trend was further illustrated in Figure 5, which showed that teams in the MOBA genre won high price amounts more frequently than expected, a major factor contributing to the observed association between game genre and earnings.

References

Daoud, J. (2020). Esports Earnings for Players, Teams, and Games. [dataset] Kaggle. Available at: https://www.kaggle.com/datasets/jackdaoud/esports-earnings-for-players-teams-by-game [Accessed 10 April 2025]

Thevapalan, A. (2024). Chi-Square Test in R: A Complete Guide, Datacap, Available online: https://www.datacamp.com/tutorial/chi-square-test-r [Accessed 1 April 2025]

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